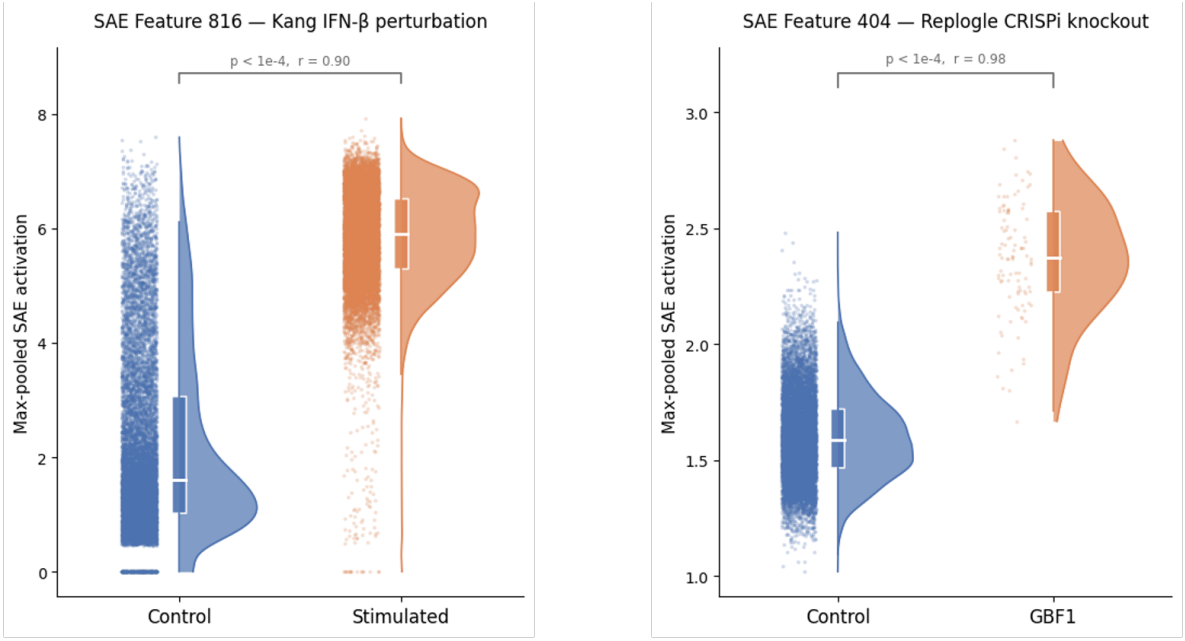

Figure 1: Validating two biological features learned by a sparse autoencoder trained on the CellxGene census. Both features show significant upregulation in perturbed cells. On the right, cells are stimulated with IFN-beta, which increases the immune response. On the left, the GBF1 gene, which is essential for the cellular secretory pathway, is knocked out, disrupting ER-to-Golgi transport. P values and rank-biserial correlations are from the Mann–Whitney U test.